

Advanced Time Series Analysis and Forecasting

Seminar 11: Spectral and wavelet analysis



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Acronyms used in this material

- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **BG**: Bulgaria (Eurostat country code)
- **BK**: Baxter–King (band-pass filter)
- **BUX**: Budapesti Értéktőzsde index (the Budapest Stock Exchange index)
- **CEE**: Central and Eastern Europe
- **CF**: Christiano–Fitzgerald (band-pass filter)
- **COI**: Cone Of Influence (of a wavelet transform)
- **CWT**: Continuous Wavelet Transform
- **CZ**: Czechia (Eurostat country code)
- **DPSS**: Discrete Prolate Spheroidal Sequences (Slepian tapers)
- **DWT**: Discrete Wavelet Transform
- **EA20**: Euro Area, 20 member states (Eurostat code)
- **EODHD**: EOD Historical Data (market-data provider)
- **FRED**: Federal Reserve Economic Data
- **GDP**: Gross Domestic Product
- **HP**: Hodrick–Prescott (filter)
- **HU**: Hungary (Eurostat country code)
- **LA**: Least Asymmetric (Daubechies wavelet filter)
- **MODWT**: Maximal Overlap Discrete Wavelet Transform
- **OLS**: Ordinary Least Squares
- **PL**: Poland (Eurostat country code)
- **PX**: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- **SK**: Slovakia (Eurostat country code)
- **US**: United States
- **VAR**: Vector AutoRegression
- **WIG20**: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

Today's question and seminar structure

- The question of the seminar
  - **how do we estimate spectra with valid bands, measure co-movement and causality by frequency, extract cycles honestly and localise co-movement in time?**
- Outputs to produce
  - the answers to the A exercises
  - one figure and one table for each B exercise
  - the analysis requested in C1 and the verdicts on the AI answer in C2, each with one line of justification
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- The exercises, their types and their order: the exercise map (next slide)
- Seminar: Prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1, A2	the HP cycle of a random walk; other frequencies and the Hamilton filter	Solved, Proposed	A1
A3, A4	bands of lag-window and multitaper estimates; resolution	Solved, Proposed	A3
A5, A6	cross-spectrum of a delayed noisy copy; a Geweke measure	Solved, Proposed	A5
A7, A8	a Haar MODWT by hand; Morlet scale, period and cone of influence	Solved, Proposed	A7
B1, B2	the spectrum of US industrial production; residual seasonality	Solved, Proposed	B1
B3, B4	coherence of output and unemployment; Breitung–Candelon	Solved, Proposed	B3
B5, B6	four filters on US GDP; real-time euro-area cycles	Solved, Proposed	B5
B7, B8	wavelet coherence of Bitcoin and S&P 500; CEE markets by scale	Solved, Proposed	B7
C1, C2	CEE business-cycle synchronisation; what is wrong in an AI answer?	Proposed	B3, A1–A8

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; the solution is discussed in the seminar
- Part A: derivations and computations on paper; Part B: real data, each exercise ending with an interpretation question; Part C: open questions and the critique of an AI answer

Data used

Series	Source	Frequency, period
US industrial production, unemployment rate	FRED	monthly, 1960–2026
US real GDP, 10-year and 3-month yields	FRED	quarterly, 1947–2026
German industrial production	Eurostat	monthly, 2000–2026
GDP: euro area, BG, CZ, HU, PL, RO, SK	Eurostat	quarterly, 1995–2026
Bitcoin, S&P 500, DAX, WIG20, BUX, PX	EODHD	daily and weekly, 2000–2026

- Growth rates are 100 times log differences
- Functions in the notebook
  - `multitaper`, `lag_window`, `cross_spectrum`, `breitung_candelon`, `hp_filter`, `bk_filter`, `hamilton_filter`, `modwt`, `wavelet_corr`, `wtc_montecarlo`
- No account or key is needed

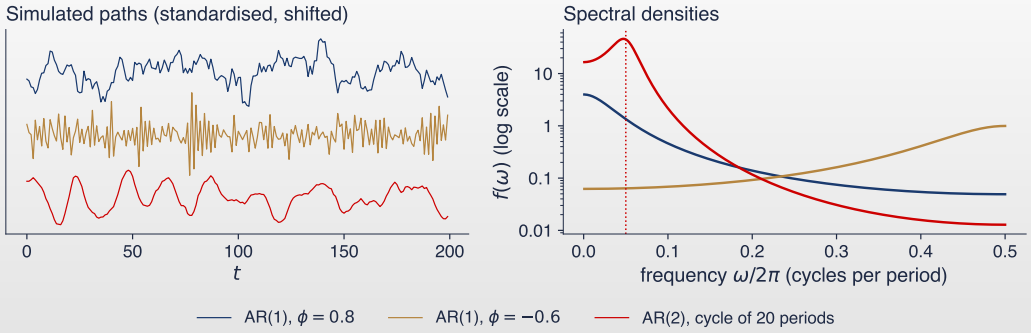
Prerequisites for Today (1/23): the spectral density

- **Spectral density** of a stationary series: the variance decomposed by frequency

$$f(\omega) = \frac{1}{2\pi} \sum_{h=-\infty}^{\infty} \gamma(h) e^{-i\omega h}, \qquad \gamma(0) = \int_{-\pi}^{\pi} f(\omega)\, d\omega$$

  - $\gamma(h) = \mathrm{Cov}(X_{t+h}, X_t)$: autocovariance at lag h; $f(\omega) = f(-\omega) \geq 0$
  - $\omega \in [-\pi, \pi]$: angular frequency in radians per period; $\omega/2\pi$: cycles per period; period $= 2\pi/\omega$
  - $\int_{\omega_1 \leq |\omega| \leq \omega_2} f$: the variance due to cycles with periods between $2\pi/\omega_2$ and $2\pi/\omega_1$
- Examples
  - white noise with variance σ^2: flat, $f = \sigma^2/(2\pi)$
  - AR(1) $X_t = \phi X_{t-1} + \varepsilon_t$: $f(\omega) = \frac{\sigma^2}{2\pi} \frac{1}{1 - 2\phi \cos\omega + \phi^2}$, peak at 0 if $\phi > 0$, at π if $\phi < 0$
  - business-cycle band: periods of 6–32 quarters (18–96 months)

Prerequisites for Today (2/23): spectra of three AR models



- $\phi = 0.8$: smooth paths, variance at low frequencies; $\phi = -0.6$: zig-zag paths, variance at high frequencies
- AR(2) with complex roots, $\phi_1 = 1.8\cos(2\pi/20)$, $\phi_2 = -0.81$: a peak at $\omega/2\pi = 1/20$, a quasi-cycle of 20 periods

Prerequisites for Today (3/23): linear filters and their gain

- A **linear filter** $Y_t = A(L)X_t = \sum_j a_j X_{t-j}$ multiplies the spectrum by the squared gain

$$f_Y(\omega) = |A(e^{-i\omega})|^2 f_X(\omega), \qquad A(e^{-i\omega}) = \sum_j a_j e^{-i\omega j}$$

  - L: lag operator, $LX_t = X_{t-1}$; $A(e^{-i\omega})$: transfer function
  - gain $|A(e^{-i\omega})|$: by how much a cycle of frequency ω is amplified (gain > 1) or damped (< 1)
  - phase shift $\arg A(e^{-i\omega})$: by how much the cycle is moved in time; zero for symmetric filters
- Example: the first difference $1 - L$
  - $|1 - e^{-i\omega}|^2 = 2(1 - \cos\omega) = 4\sin^2(\omega/2)$: removes $\omega = 0$, doubles the amplitude at $\omega = \pi$
  - a random walk $\Delta y_t = \varepsilon_t$ has the pseudo-spectrum $f_y = f_{\Delta y}/(2(1 - \cos\omega))$, infinite at 0

Prerequisites for Today (4/23): the periodogram

- **Periodogram** at the Fourier frequencies: the squared modulus of the discrete Fourier transform

$$I(\omega_j) = \frac{1}{2\pi n}\left|\sum_{t=1}^{n}(X_t - \bar{X})e^{-i\omega_j t}\right|^2, \qquad \omega_j = \frac{2\pi j}{n}, \quad j = 1, \ldots, \lfloor n/2 \rfloor$$

  - n: sample size; $\bar{X}$: sample mean; j cycles fit exactly in the sample at ω_j

- Its distribution

$$I(\omega_j) \approx f(\omega_j)\frac{\chi^2_2}{2}, \qquad \mathbb{E}I(\omega_j) \to f(\omega_j), \qquad \mathrm{Var}I(\omega_j) \to f(\omega_j)^2$$

  - χ^2_ν: chi-square distribution with ν degrees of freedom (sum of ν squared independent $N(0,1)$), mean ν, variance 2ν
  - unbiased in the limit, but **inconsistent**: the variance does not shrink as n grows
  - ordinates at different j are nearly independent: averaging neighbours reduces the variance

Prerequisites for Today (5/23): lag-window estimators

- **Lag window**: down-weight the sample autocovariances at long lags

$$\hat{f}(\omega) = \frac{1}{2\pi} \sum_{|h|<n} k\Big(\frac{h}{M}\Big)\hat{\gamma}(h)e^{-i\omega h}, \qquad \hat{\gamma}(h) = \frac{1}{n}\sum_{t=1}^{n-h}(X_{t+h} - \bar{X})(X_t - \bar{X})$$

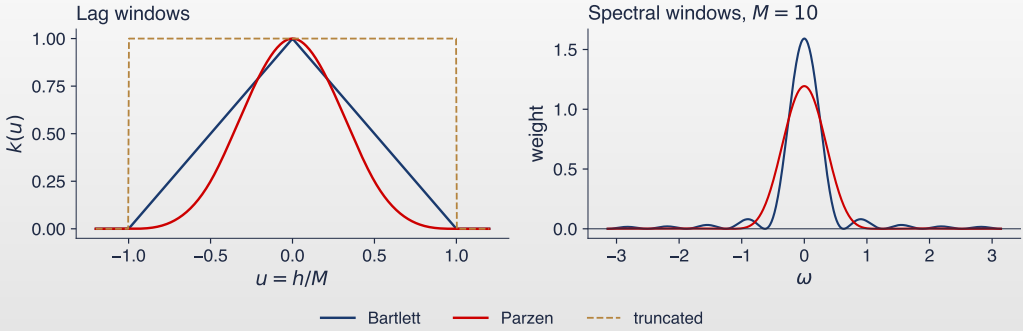
  - M: bandwidth (truncation point); $k(u)$: kernel, $k(0) = 1$, $k(u) = 0$ for $|u| > 1$
  - Bartlett: $k(u) = 1 - |u|$; Parzen: $1 - 6u^2 + 6|u|^3$ for $|u| \le 1/2$, $2(1 - |u|)^3$ for $1/2 < |u| \le 1$

- **Confidence band**: $\nu\hat{f}/f$ is approximately χ^2_ν

$$\nu = \frac{2n}{M\int_{-1}^{1} k^2(u)\,du}, \qquad \Big[\frac{\nu\hat{f}(\omega)}{\chi^2_{\nu,0.975}}, \frac{\nu\hat{f}(\omega)}{\chi^2_{\nu,0.025}}\Big]$$

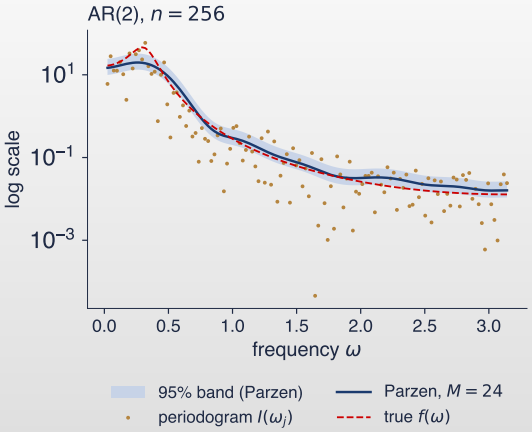
  - ν: equivalent degrees of freedom; $\int k^2 = 2/3$ (Bartlett), 0.539 (Parzen); $\chi^2_{\nu,p}$: quantile of order p
  - larger M: less bias (sharper peaks), more variance (fewer degrees of freedom)

Prerequisites for Today (6/23): lag windows and spectral windows



- Left: the weights $k(h/M)$ on $\hat{\gamma}(h)$; right: the equivalent weights on the periodogram around ω (spectral window)
- The spectral window has width of order $1/M$: $\hat{f}(\omega)$ is a local average of I over that width, which blurs sharp peaks (bias)

Prerequisites for Today (7/23): smoothing the periodogram



- Simulated AR(2) with a 20-period cycle, $n = 256$
- The periodogram scatters over three orders of magnitude around f
  - one $\chi^2_2/2$ draw per frequency
- Parzen with $M = 24$: $\nu = 39.6$ degrees of freedom
  - the band is a constant factor 2.44 wide (upper over lower end)
  - on a log scale it has the same width everywhere
- At the peak the estimate is too low: the band ignores this bias

Prerequisites for Today (8/23): multitaper estimation

- **Multitaper** Thomson (1982): K periodograms of the data multiplied by orthogonal tapers $h_t^{(k)}$, then averaged

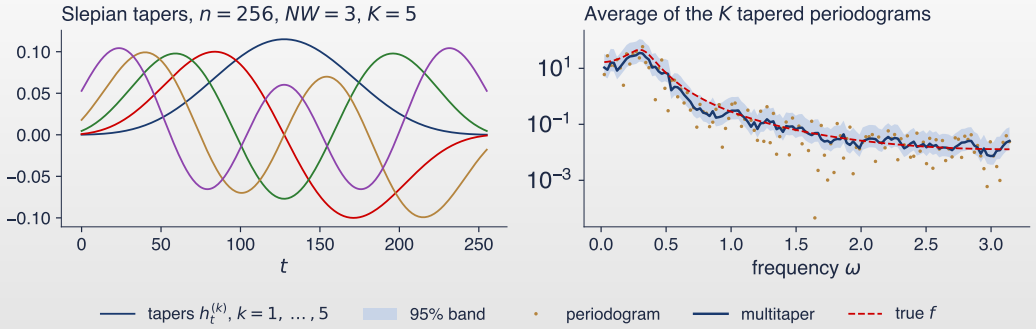
$$\hat{S}_k(\omega) = \frac{1}{2\pi}\Big|\sum_{t=1}^{n} h_t^{(k)} X_t e^{-i\omega t}\Big|^2, \qquad \hat{f}(\omega) = \frac{1}{K}\sum_{k=1}^{K}\hat{S}_k(\omega), \qquad \frac{2K\hat{f}}{f} \approx \chi^2_{2K}$$

  - taper: a weight sequence with $\sum_t h_t^2 = 1$ that fades the data at the ends (less leakage from strong peaks)
  - Slepian (DPSS) tapers: the most concentrated in the band $[\omega - 2\pi W, \omega + 2\pi W]$; $K = 2NW - 1$ of them

- Resolution and variance
  - NW: time–bandwidth product; $W = NW/n$: half-bandwidth in cycles per observation; two peaks closer than $2W$ merge
  - 95% band: $[2K\hat{f}/\chi^2_{2K,0.975},\ 2K\hat{f}/\chi^2_{2K,0.025}]$; larger NW: more tapers, less variance, coarser resolution

- **Andrews bandwidth** Andrews (1991) (B1): an AR(1) plug-in rule designed for $f(0)$
  - Parzen: $M = 2.6614(\hat{\alpha}n)^{1/5}$, $\hat{\alpha} = 4\hat{\rho}^2/(1-\hat{\rho})^4$, $\hat{\rho}$: lag-1 autocorrelation

Prerequisites for Today (9/23): Slepian tapers



- Left: the $K = 5$ tapers for $NW = 3$; each keeps at least 0.946 of its energy inside the band $\pm W$
- Right: the same AR(2) as before; the average of 5 tapered periodograms has 10 degrees of freedom and a narrow band

Prerequisites for Today (10/23): the harmonic F test

- **Line component**: a deterministic cycle $\mu\cos(\omega_0 t + \theta)$ added to a series with a smooth spectrum
  - seasonal and calendar effects are lines at fixed frequencies (e.g. $k/12$ cycles per month, $k = 1, \ldots, 6$)
- **Thomson's F test** Thomson (1982): regress the tapered transforms $Y_k(\omega)$ on $U_k = \sum_t h_t^{(k)}$

$$\hat{\mu}(\omega) = \frac{\sum_k U_k Y_k(\omega)}{\sum_k U_k^2}, \qquad F(\omega) = \frac{(K-1)|\hat{\mu}(\omega)|^2 \sum_k U_k^2}{\sum_k |Y_k(\omega) - \hat{\mu}(\omega) U_k|^2} \sim F(2, 2K-2)$$

  - $Y_k(\omega) = \sum_t h_t^{(k)} X_t e^{-i\omega t}$; numerator: power explained by a line; denominator: the remaining (background) power
  - $F(a, b)$: Fisher distribution; a large F (small p-value): a line at ω
- Multiple testing
  - over N frequencies about αN tests reject at level α by chance: test pre-specified frequencies

Prerequisites for Today (11/23): cross-spectrum, coherence, phase and gain

- **Cross-spectrum** of two series: the Fourier transform of the cross-covariances

$$f_{xy}(\omega) = \frac{1}{2\pi} \sum_h \gamma_{xy}(h) e^{-i\omega h} = c(\omega) - i\, q(\omega), \qquad \gamma_{xy}(h) = \mathrm{Cov}(x_{t+h}, y_t)$$

  - complex-valued; c: co-spectrum (in-phase covariance), q: quadrature spectrum (covariance shifted by a quarter cycle)

- Three summaries at each frequency

$$\kappa^2(\omega) = \frac{|f_{xy}(\omega)|^2}{f_x(\omega) f_y(\omega)} \in [0, 1], \qquad \phi(\omega) = \arg f_{xy}(\omega), \qquad \text{gain} = \frac{|f_{xy}(\omega)|}{f_x(\omega)}$$

  - squared coherence κ^2: an R^2 by frequency, the share of the variance of y at ω explained linearly by x
  - phase $\phi \in (-\pi, \pi]$: if $y_t = x_{t-d}$, $\phi = \omega d$ and x leads by ϕ/ω periods; $\phi = \pi$: anti-phase (opposite signs)
  - gain: the slope of the regression of y on x at frequency ω

Prerequisites for Today (12/23): inference on coherence and phase

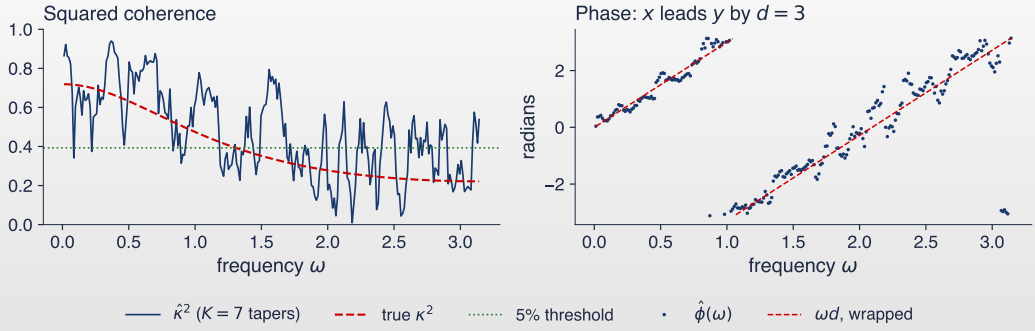
- **Threshold**: with L averaged ordinates (tapers), under zero coherence

$$P(\hat{\kappa}^2 > c) = (1-c)^{L-1} \quad \Rightarrow \quad c_{5\%} = 1 - 0.05^{1/(L-1)}$$

  - read the phase only where $\hat{\kappa}^2$ exceeds the threshold: with no coherence the phase is pure noise
- **Lead in periods** from a phase near anti-phase: the deviation $\phi - \pi$ divided by ω
  - circular mean of angles ϕ_j: $\arg \sum_j e^{i\phi_j}$ (an ordinary mean fails near $\pm\pi$)
  - Okun's law (B3): unemployment changes move opposite to output growth, so the expected phase is π
- **Dynamic correlation** Croux, Forni and Reichlin (2001) over a band Ω: a correlation with a sign, unlike coherence

$$\rho(\Omega) = \frac{\int_\Omega c(\omega)\, d\omega}{\sqrt{\int_\Omega f_x(\omega)\, d\omega \int_\Omega f_y(\omega)\, d\omega}} \in [-1, 1]$$

Prerequisites for Today (13/23): coherence and phase of a delayed copy



- Simulated: x_t an AR(1) with $\phi = 0.5$, $y_t = 0.8x_{t-3} + u_t$, u_t white noise with variance 1, $n = 512$
- Coherence is high where x has much power (low ω); 5% threshold 0.39; the phase follows 3ω, wrapped at $\pm\pi$

Prerequisites for Today (14/23): causality by frequency

- **VAR(p) of $Z_t = (x_t, y_t)'$**: $Z_t = \sum_{j=1}^{p} A_j Z_{t-j} + e_t$, $\text{Var} e_t = \Sigma$
  - transfer function $H(\omega) = (I - \sum_j A_j e^{-ij\omega})^{-1}$; spectral matrix $\frac{1}{2\pi} H \Sigma H^*$ (*: conjugate transpose)
  - Granger causality $y \to x$: past y improves the forecast of x; ψ_j: coefficient of y_{t-j} in the equation of x
- **Geweke measure** Geweke (1982): how much of $f_x(\omega)$ comes from y

$$M_{y\to x}(\omega) = \ln \frac{2\pi f_x(\omega)}{|\tilde{H}_{xx}(\omega)|^2 \tilde{\sigma}_{xx}} \geq 0, \qquad \frac{1}{\pi}\int_0^\pi M_{y\to x}(\omega)\, d\omega = \ln \frac{\sigma^2_{x|x}}{\sigma^2_{x|x,y}}$$

  - $\tilde{H}_{xx}$, $\tilde{\sigma}_{xx}$: the transfer function and variance of x's own innovation after orthogonalising e_t; $M = 0$: y does not help at ω
  - under regularity conditions the average over frequencies is the usual Granger measure (σ^2: one-step forecast-error variances)
- **Breitung–Candelon test** Breitung and Candelon (2006): $M_{y\to x}(\omega) = 0$ is two linear restrictions

$$H_0 : \sum_{j=1}^{p} \psi_j \cos(j\omega) = \sum_{j=1}^{p} \psi_j \sin(j\omega) = 0, \qquad F \sim F(2, T-2p-1)$$

  - T: number of observations; VAR order by AIC: $\min_p \ln|\hat{\Sigma}_p| + 2pk^2/T$ ($k = 2$ series)

Prerequisites for Today (15/23): the Hodrick–Prescott filter

- **HP** Hodrick and Prescott (1997): the trend that balances fit and smoothness

$$\hat{\tau} = \arg\min_{\tau} \sum_{t=1}^{n} (y_t - \tau_t)^2 + \lambda \sum_{t=2}^{n-1} (\Delta^2 \tau_{t+1})^2 = (I + \lambda D' D)^{-1} y$$

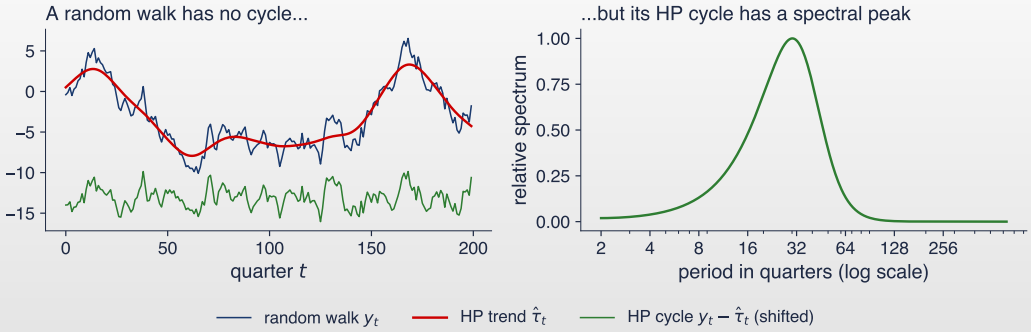
  - $\Delta^2 \tau_{t+1} = \tau_{t+1} - 2\tau_t + \tau_{t-1}$; D: the $(n-2) \times n$ second-difference matrix; cycle $c_t = y_t - \hat{\tau}_t$
  - λ: smoothness penalty; $\lambda \to \infty$: a linear trend; $\lambda = 0$: $\hat{\tau} = y$

- Gain of the cycle filter (from y to c), in the middle of a long sample

$$G(\omega) = \frac{4\lambda(1 - \cos\omega)^2}{1 + 4\lambda(1 - \cos\omega)^2}$$

  - a high-pass filter: $G \approx 1$ for short cycles, $\to 0$ for long ones; $\lambda = 1600$ (quarters): $G = 1/2$ near 40 quarters
  - Ravn–Uhlig Ravn and Uhlig (2002): λ scales with the fourth power of the observation frequency: 129 600 (months), 6.25 (years)

Prerequisites for Today (16/23): a cycle made by the filter



- A simulated random walk has no cycle: its spectrum decreases monotonically from $\omega = 0$
- Its HP cycle has the spectrum $G(\omega)^2 f_y(\omega)$, the product of a rising gain and a falling spectrum: a peak appears Cogley and Nason (1995)

Prerequisites for Today (17/23): band-pass and regression filters

- **Ideal band-pass** for periods $[p_l, p_u]$: gain 1 for $\omega_{\mathbf{1}} = 2\pi/p_u \le \omega \le \omega_{\mathbf{2}} = 2\pi/p_l$, else 0

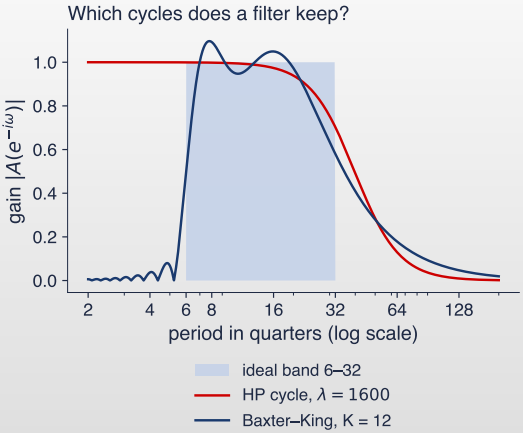
$$b_{\mathbf{0}} = \frac{\omega_{\mathbf{2}} - \omega_{\mathbf{1}}}{\pi}, \qquad b_j = \frac{\sin(j\omega_{\mathbf{2}}) - \sin(j\omega_{\mathbf{1}})}{\pi j}, \quad j = \pm 1, \pm 2, \ldots$$

  - infinitely many weights: any real filter approximates them
- **Baxter–King** Baxter and King (1999): keep $j = -K, \ldots, K$ and add a constant so that $\sum_j a_j = 0$
  - $a_j = b_j + \theta$, $\theta = -\sum_{|j| \le K} b_j/(2K+1)$; zero sum removes a unit root; K observations lost at each end
  - **Christiano–Fitzgerald** Christiano and Fitzgerald (2003): asymmetric weights that use the whole sample, no observations lost
- **Hamilton** Hamilton (2018): the cycle is the error of an 8-quarter-ahead OLS forecast

$$y_{t+\mathbf{8}} = \beta_{\mathbf{0}} + \beta_{\mathbf{1}} y_t + \beta_{\mathbf{2}} y_{t-\mathbf{1}} + \beta_{\mathbf{3}} y_{t-\mathbf{2}} + \beta_{\mathbf{4}} y_{t-\mathbf{3}} + v_{t+\mathbf{8}}, \qquad c_{t+\mathbf{8}} = \hat{v}_{t+\mathbf{8}}$$

  - for a random walk $\beta_{\mathbf{1}} = 1$, others 0: the cycle is $y_{t+\mathbf{8}} - y_t$

Prerequisites for Today (18/23): filter gains and real time



- Gains against the period (quarters)
  - BK: close to the ideal band, ripples at the edges
  - HP: keeps everything shorter than 32 quarters and part of the longer cycles (gain 0.70 at 32)
- **Real time** against **final**
  - real time: the cycle at t computed with data up to t only (one-sided); final: with the whole sample
  - revision $c_t^{\text{final}} - c_t^{\text{rt}}$; root mean square revision $\sqrt{\frac{1}{T}\sum_t (c_t^{\text{final}} - c_t^{\text{rt}})^2}$
  - share of sign revisions: quarters where the two cycles have opposite signs

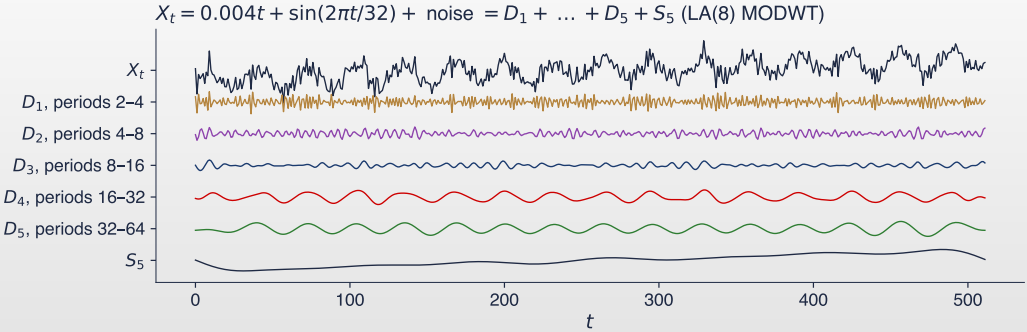
Prerequisites for Today (19/23): the MODWT

- **MODWT** Percival and Walden (2000): filter the series with the level-j wavelet filter, circularly

$$\tilde{W}_{j,t} = \sum_{l=0}^{L_j-1} \tilde{h}_{j,l} X_{(t-l) \bmod n}, \qquad \|X\|^2 = \sum_{j=1}^{J} \|\tilde{W}_j\|^2 + \|\tilde{V}_J\|^2$$

  - level j: changes over periods $[2^j, 2^{j+1}]$; $\tilde{V}_J$: scaling (smooth) coefficients, periods above 2^{J+1}; L_j: filter length
  - Haar, level 1: $\tilde{W}_{1,t} = (X_t - X_{t-1})/2$, $\tilde{V}_{1,t} = (X_t + X_{t-1})/2$; LA(8): a smoother filter of length 8
  - the first $L_j - 1$ coefficients wrap around the end of the sample (boundary): they are dropped; any n works
- Statistics by level, from the M_j non-boundary coefficients
  - wavelet variance $\hat{\nu}_j^2 = \frac{1}{M_j} \sum_t \tilde{W}_{j,t}^2$: the variance of X at scale j
  - wavelet correlation $\hat{\rho}_j = \dfrac{\sum_t \tilde{W}^X_{j,t} \tilde{W}^Y_{j,t}}{\sqrt{\sum_t (\tilde{W}^X_{j,t})^2 \sum_t (\tilde{W}^Y_{j,t})^2}}$; wavelet beta $\hat{\beta}_j = \dfrac{\sum_t \tilde{W}^X_{j,t} \tilde{W}^Y_{j,t}}{\sum_t (\tilde{W}^X_{j,t})^2}$
  - 95% Fisher-z interval Whitcher, Guttorp and Percival (2000): $\tanh\left(\operatorname{atanh} \hat{\rho}_j \pm 1.96/\sqrt{M_j/2^j - 3}\right)$

Prerequisites for Today (20/23): a multiresolution analysis



- The details D_j (the MODWT of level j mapped back to time) and the smooth S_5 add up exactly to X_t
- The 32-period sine lies at the border of D_4 and D_5; the noise sits in D_1–D_2; the trend in S_5

Prerequisites for Today (21/23): the continuous wavelet transform

- **Morlet CWT** Torrence and Compo (1998): correlate the series with shifted and stretched copies of a wave packet

$$\psi(u) = \pi^{-1/4} e^{i\omega_0 u} e^{-u^2/2}, \qquad W(s,t) = \sum_{k=1}^{n} X_k \frac{1}{\sqrt{s}} \psi^*\Big(\frac{k-t}{s}\Big)$$

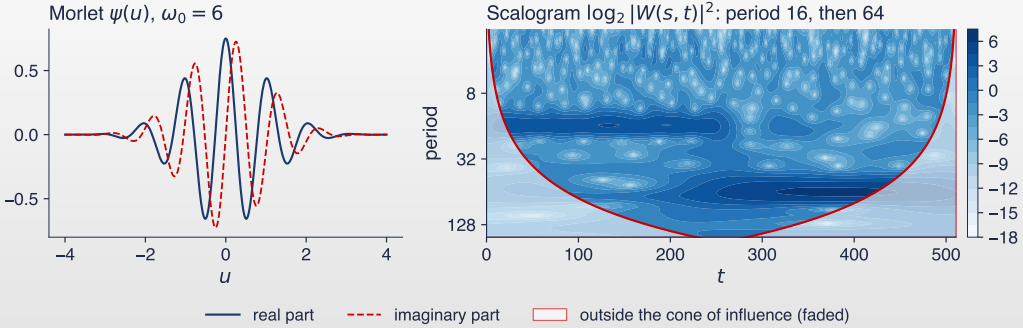
  - s: scale (stretch); t: location in time; ψ^*: complex conjugate; $\omega_0 = 6$: number of oscillations in the packet
  - $|W(s,t)|^2$: local power of cycles of scale s around time t (scalogram); scales $s_j = s_0 2^{j\,\delta j}$, δj: spacing in octaves

- Scale, period and the cone of influence

$$\text{period} = \frac{4\pi}{\omega_0 + \sqrt{2+\omega_0^2}}\, s \approx 1.033\, s, \qquad \text{reliable if } \min(t, n-t) > \sqrt{2}\, s$$

  - cone of influence (COI): within $\sqrt{2}\, s$ of either end the packet runs off the sample and $|W|^2$ is damped

Prerequisites for Today (22/23): a Morlet scalogram



- Left: the Morlet packet, a sine wave inside a Gaussian envelope
- Right: a simulated series with period 16 in the first half and 64 in the second: the CWT locates both in time, a spectrum would only show two peaks; outside the red curve (COI) do not interpret

Prerequisites for Today (23/23): wavelet coherence

- **Wavelet coherence** Grinsted, Moore and Jevrejeva (2004): a local R^2 in time and scale

$$R^2(s,t) = \frac{|\mathcal{S}(s^{-1}W^{xy}(s,t))|^2}{\mathcal{S}(s^{-1}|W^x(s,t)|^2)\,\mathcal{S}(s^{-1}|W^y(s,t)|^2)} \in [0,1]$$

  - W^x, W^y: the CWTs; cross-wavelet $W^{xy} = W^x W^{y*}$; $\mathcal{S}$: smoothing in time (Gaussian, width s) and across scales
  - phase $\arg \mathcal{S}(s^{-1}W^{xy})$ as arrows: right in phase, left anti-phase, up x leads

- **Significance by Monte Carlo**
  - simulate many pairs of independent AR(1) series with the lag-1 autocorrelations of x and y; the 95th percentile of R^2 at each scale is the 5% contour
  - pointwise tests: about 5% of the reliable area is significant by chance; compare the observed share with its null distribution

A1: the spurious cycle of the HP filter [Solved]

Task

- y_t is a random walk, $\Delta y_t = \varepsilon_t$ white noise; HP with $\lambda = 1600$. This replicates Cogley and Nason (1995).
- 1. Write the spectrum of the HP cycle c_t using the gain $G(\omega)$ and $f_{\Delta y}$.
- 2. With $u = 1 - \cos\omega$, find the u^* that maximises it, then the peak frequency and period.
- 3. Compute the ratio of the spectrum at a period of 8 quarters to its peak.
- Report: one formula; three numbers; one number and one sentence.

Solution

- 1. $f_y = f_{\Delta y}/|1 - e^{-i\omega}|^2 = \frac{\sigma^2}{2\pi \cdot 2u}$, so $f_c = G^2 f_y \propto \frac{(4\lambda u^2)^2}{(1+4\lambda u^2)^2\, u} \propto \frac{u^3}{(1+4\lambda u^2)^2}$
- 2. Derivative zero: $3(1 + 4\lambda u^2) = 16\lambda u^2$, $u^* = \sqrt{3/(4\lambda)} = 0.0217$; $\omega^* = \arccos(1 - u^*) = 0.2085$; period $2\pi/\omega^* = 30.1$ quarters
- 3. Ratio 0.13
- Interpretation: a series with no cycle at all gets, after HP filtering, a dominant cycle of about 7.5 years: the "business cycle" can be manufactured by the filter

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A2: other frequencies and the Hamilton filter [Proposed]

Solution discussed in class

Task

- Random walk as in A1. Model: A1.
- 1. Repeat A1 for monthly data with $\lambda = 129\,600$ and annual data with $\lambda = 6.25$; express both peaks in years.
- 2. For Hamilton's filter with $h = 8$, the random-walk cycle is $y_{t+8} - y_t = \sum_{k=1}^{8} \varepsilon_{t+k}$: write its spectrum and find its zeros and its maximum.
- 3. Does the Hamilton filter create a spurious peak inside the business-cycle band?
- Report: four numbers; one formula and two sets of numbers; one sentence.

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A3: confidence bands of lag-window estimates [Solved]

Task

- $n = 200$, truncation $M = 20$, estimate $\hat{f}(\omega) = 2.0$ at an interior frequency.
- 1. Compute the equivalent degrees of freedom ν for Bartlett and for Parzen.
- 2. Build the 95% band for each.
- 3. Why is the band asymmetric, and what does it ignore?
- Report: two numbers; two intervals; two sentences.

Solution

- 1. Bartlett $\nu = 400/(20 \cdot 2/3) = 30.0$; Parzen $\nu = 400/(20 \cdot 0.539) = 37.1$
- 2. Bartlett [1.28, 3.57]; Parzen [1.33, 3.35]
- 3. χ^2 is skewed; on a log scale the band is symmetric with constant width; it ignores bias: near a peak the true f can lie outside the band
- Interpretation: at the same M Parzen has more degrees of freedom (less variance) but a wider spectral window (more bias)

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A4: multitaper bands and resolution [Proposed]

Solution discussed in class

Task

- $n = 256$, $NW = 3$; an estimate $\hat{f}(\omega) = 0.8$. Model: A3.
- 1. Give K, the degrees of freedom and the half-bandwidth W in cycles per observation.
- 2. Build the 95% band.
- 3. Can two spectral peaks at 0.100 and 0.115 cycles be resolved? If not, what NW or what n would resolve them?
- Report: three numbers; one interval; one sentence and two numbers.

A5: the cross-spectrum of a delayed noisy copy [Solved]

Task

- x_t white noise with variance 1; $y_t = 0.8x_{t-2} + u_t$, u_t white noise with variance 0.5, independent of x.
- 1. Compute $f_{xy}(\omega)$ with $\gamma_{xy}(h) = \mathrm{Cov}(x_{t+h}, y_t)$.
- 2. Compute the squared coherence, the phase at $\omega = \pi/4$ and the gain.
- 3. With $K = 7$ tapers, is a sample coherence equal to the true one significant at 5%?
- Report: one formula; three numbers; one number and one sentence.

Solution

- 1. $\mathrm{Cov}(x_{t+h}, 0.8x_{t-2}) = 0.8$ only at $h = -2$, so $f_{xy}(\omega) = \frac{0.8}{2\pi} e^{2i\omega}$
- 2. $\kappa^2 = 0.64/(0.64 + 0.5) = 0.561$ at every ω; phase 2ω, here $\pi/2 = 1.571$: x leads y by $\phi/\omega = 2$ periods; gain 0.8
- 3. Threshold $1 - 0.05^{1/6} = 0.393 < 0.561$: yes
- Interpretation: coherence measures the noise share, phase the delay, gain the slope: three different pieces of information

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A6: a Geweke measure by hand [Proposed]

Solution discussed in class

Task

- VAR(1): $x_t = 0.5x_{t-1} + 0.4y_{t-1} + e_{1t}$, $y_t = 0.6y_{t-1} + e_{2t}$, $\Sigma = I$. Model: A5.
- 1. Write x_t as a filter of e_1 and e_2 and show $M_{y\to x}(\omega) = \ln\left(1 + 0.16/|1 - 0.6e^{-i\omega}|^2\right)$.
- 2. Evaluate it at $\omega = 0, \pi/2, \pi$ and its average $\frac{1}{\pi}\int_0^\pi M$.
- 3. Why can the Breitung–Candelon test not distinguish frequencies in this VAR(1)?
- Report: one derivation; four numbers; one sentence.

A7: a Haar MODWT by hand [Solved]

Task

- $x = (2, 4, 3, 7, 6, 5, 9, 8)$; Haar MODWT, one level, circular ($x_{-\mathbf{1}} = x_{\mathbf{7}}$).
- 1. Compute the Haar coefficients $\tilde{W}_{\mathbf{1},t}$ and $\tilde{V}_{\mathbf{1},t}$ for $t = 0, \ldots, 7$.
- 2. Check the energy identity $\|x\|^{\mathbf{2}} = \|\tilde{W}_{\mathbf{1}}\|^{\mathbf{2}} + \|\tilde{V}_{\mathbf{1}}\|^{\mathbf{2}}$.
- 3. Estimate the level-1 wavelet variance without the boundary coefficient $t = 0$.
- Report: sixteen numbers; three numbers; one number.

Solution

- 1. $\tilde{W}_{\mathbf{1}} = (-3.0;\ 1.0;\ -0.5;\ 2.0;\ -0.5;\ -0.5;\ 2.0;\ -0.5)$; $\tilde{V}_{\mathbf{1}} = (5.0;\ 3.0;\ 3.5;\ 5.0;\ 6.5;\ 5.5;\ 7.0;\ 8.5)$
- 2. $\|x\|^{\mathbf{2}} = 284 = 19 + 265$
- 3. $\hat{\nu}^{\mathbf{2}}_{\mathbf{1}} = \frac{\mathbf{1}}{\mathbf{7}} \sum_{t=\mathbf{1}}^{\mathbf{7}} \tilde{W}^{\mathbf{2}}_{\mathbf{1},t} = 1.429$
- Interpretation: $\tilde{W}_{\mathbf{1},\mathbf{0}} = (2 - 8)/2$ mixes the first and the last observation: that is why boundary coefficients are dropped; the MODWT works for $n = 8$ as for any n

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A8: Morlet scale, period and cone of influence [Proposed]

Solution discussed in class

Task

- Weekly data, $n = 520$ (ten years); Morlet $\omega_0 = 6$. Model: A7.
- 1. Compute the Fourier factor (period over scale) and the scale of a 52-week period.
- 2. The cone of influence is $\sqrt{2}\,s$ at each end: what share of the sample is reliable at the 52-week period?
- 3. What is the longest period with at least one reliable point?
- Report: two numbers; one number; one number.

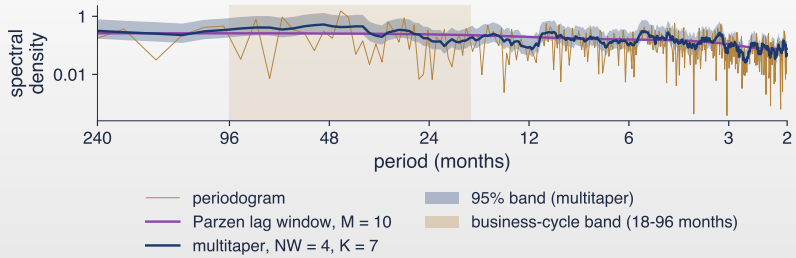
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B1: the spectrum of US industrial production [Solved]

- **Question**: where is the variance of monthly US industrial production growth, and do three estimators agree?
- **Inputs**: FRED INDPRO, monthly growth (100 $\times$ log difference), February 1960 – August 2026 ($n = 799$)
- **Tasks**
  1. Compute the periodogram, the Parzen estimate with the Andrews (AR(1) plug-in) bandwidth and the multitaper estimate with $NW = 4$ and its 95% band.
  2. Compute the share of variance in the business-cycle band (18–96 months) and the period of the highest multitaper peak below 120 months.
  3. Compare the Parzen and multitaper estimates around the peak.
  4. Interpretation: which estimate would you report, and why?
- **Report**: one chart, two numbers, one comparison, one sentence
- Notebook: section B1

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B1: solution [Solved]



- Business-cycle band: 17.5% of the variance (periodogram: 18.1%); periods above 96 months: 4.0%; highest multitaper peak near 50 months
- Andrews gives $M = 10$: the Parzen estimate is almost flat and erases the 4-year peak; the multitaper band is a factor 4.6 wide (14 degrees of freedom)
- Interpretation: report the multitaper estimate with its band; the Andrews rule is designed for the long-run variance at $\omega = 0$, not for a peak

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B2: residual seasonality in German industry [Proposed]

- **Question**: does the official seasonal adjustment of German industrial production leave seasonal or calendar lines?
- **Inputs**: Eurostat, German industrial production (B–D), unadjusted and seasonally and calendar adjusted, monthly growth 2000–2026; model: B1
- **Tasks**
  1. Run Thomson's harmonic F test ($NW = 4$, $K = 7$) on both series.
  2. Report the p-values at the six seasonal frequencies $k/12$ and at the trading-day frequency 0.348 cycles per month.
  3. Count how many other frequencies are significant at 1% and compare with the number expected by chance.
  4. Interpretation: does the adjustment pass this check?
- **Report**: two sets of seven p-values, two counts, one sentence
- Notebook: section B2

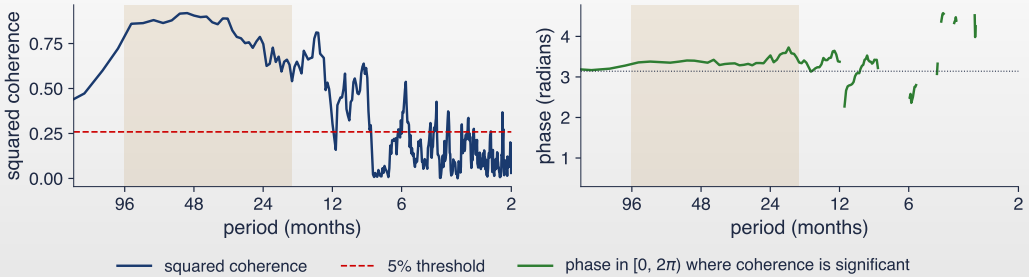
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B3: output and unemployment by frequency [Solved]

- **Question**: at which frequencies do US industrial production and unemployment move together, and which leads?
- **Inputs**: FRED INDPRO growth and the monthly change of UNRATE, 1960–2019 ($n = 719$; 2020 excluded, since April 2020 alone dominates); multitaper $NW = 6$, $K = 11$
- **Tasks**
  1. Estimate the squared coherence and its 5% threshold.
  2. Estimate the phase where coherence is significant and convert the deviation from anti-phase into a lead in months.
  3. Compare the coherence in the business-cycle band with that at periods under 6 months.
  4. Interpretation: is Okun's relation a business-cycle phenomenon?
- **Report**: one chart, three numbers, one lead in months, one sentence
- Notebook: section B3

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B3: solution [Solved]



- Coherence 0.79 in the business-cycle band, 0.14 below 6 months; threshold 0.26; simple correlation -0.40
- Phase near π (circular mean -2.87 rad): anti-phase; the deviation from π implies that production leads unemployment by about 1.3 months
- Interpretation: yes: the link is strong at business-cycle frequencies and weak month to month; a monthly regression mixes both and understates it

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B4: the yield spread and growth by frequency [Proposed]

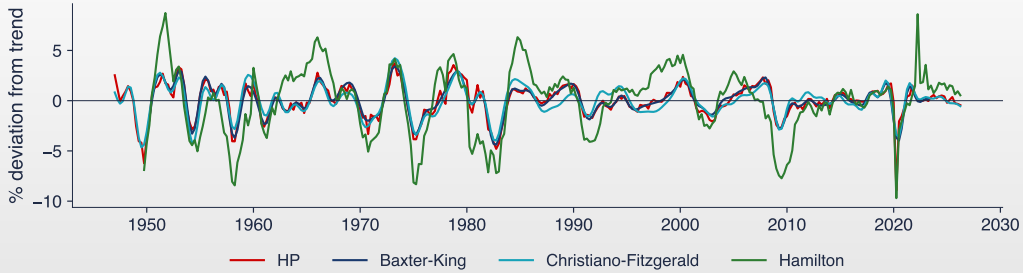
- **Question**: does the yield spread predict US output growth, and at which frequencies? Breitung and Candelon (2006) report predictive power around one year and at business-cycle frequencies
- **Inputs**: quarterly: growth = 100 × log difference of GDPC1; spread = GS10 − TB3MS (quarterly averages); 1960–2026; model: B3
- **Tasks**
  1. Choose the VAR order by AIC (at least 3) and run the Breitung–Candelon test from spread to growth over a grid of frequencies.
  2. Report the range of periods where H_0 is rejected at 5% and the p-values at 4, 8, 16 and 32 quarters.
  3. Repeat on 1960–2019.
  4. Interpretation: which part of the published result survives on data to 2026?
- **Report**: one order, one range, four p-values, the same for the shorter sample, one sentence
- Notebook: section B4

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B5: four filters on US GDP [Solved]

- **Question**: do HP, Baxter–King, Christiano–Fitzgerald and Hamilton tell the same story about the US business cycle?
- **Inputs**: 100 $\times$ log real GDP (FRED GDPC1), 1947Q1 – 2026Q2 ($n = 318$); HP $\lambda = 1600$; BK(6, 32, 12); CF(6, 32); Hamilton $h = 8$, $p = 4$ as in Hamilton (2018)
- **Tasks**
  1. Compute the four cycles and their standard deviations.
  2. Compute the pairwise correlations and the troughs in 2008–2010 and in 2020.
  3. Report the Hamilton coefficients and their sum.
  4. Interpretation: what explains the differences?
- **Report**: four numbers, a correlation table, eight troughs, five coefficients, one sentence
- Notebook: section B5

B5: solution [Solved]



- Standard deviations: HP 1.62, BK 1.48, CF 1.50, Hamilton 3.25; correlations BK–HP 0.94, HP–Hamilton 0.73, CF–Hamilton 0.64
- Troughs 2008–2010: HP -2.8, BK -2.8, Hamilton -7.7; 2020: HP -8.9, BK -4.0, Hamilton -9.7; Hamilton $\hat{\beta}_{\mathbf{1}} = 0.894$, $\sum \hat{\beta}_j = 0.979$
- Interpretation: the coefficients sum to almost one (a near random walk), so the Hamilton cycle is close to the 8-quarter growth: it keeps long cycles and doubles the 16-quarter ones; BK removes the 2020 spike, which is shorter than 6 quarters

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B6: the euro-area output gap in real time [Proposed]

- **Question**: how much would a real-time analyst have revised the euro-area cycle?
- **Inputs**: Eurostat, real GDP of the euro area (EA20), 1995Q1 – 2026Q2; first real-time estimate after 40 quarters; model: B5
- **Tasks**
  1. Compute the final and the real-time HP and Hamilton cycles.
  2. Report the root mean square revision and the share of quarters whose sign is revised, for both filters.
  3. Compare the revisions with the standard deviations of the final cycles.
  4. Interpretation: which filter would you use in real time?
- **Report**: four numbers, two standard deviations, one sentence
- Notebook: section B6

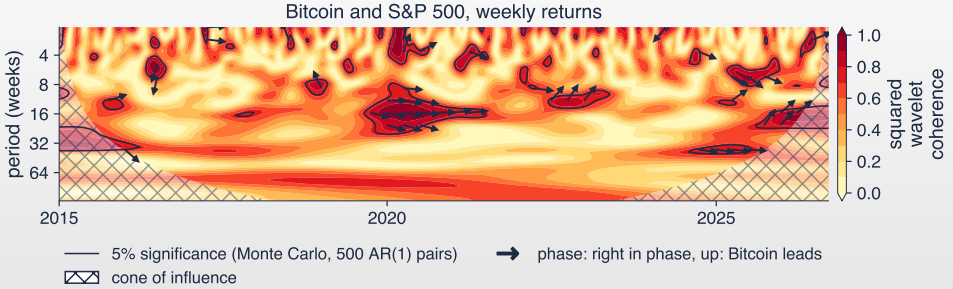
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B7: Bitcoin and the S&P 500 in time and frequency [Solved]

- **Question**: has Bitcoin become coupled with equities, and at which horizons?
- **Inputs**: EODHD weekly returns (Friday closes), January 2015 – September 2026 ($n = 611$); Morlet, $\delta j = 1/12$; 500 AR(1) pairs
- **Tasks**
  1. Compute the wavelet coherence and the Monte Carlo 5% contour.
  2. Report the share of the reliable area that is significant and its 95th percentile under the null.
  3. Compare the mean R^2 at 8–64 weeks in 2015–2019 and 2022–2026 with the simple correlations of the two periods.
  4. Interpretation: is there evidence of coupling?
- **Report**: one chart, two shares, four numbers, one sentence
- Notebook: section B7

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B7: solution [Solved]



- Significant area 7.2%, null 95th percentile 8.5%: overall, not distinguishable from independent series
- Mean R^2 at 8–64 weeks: 0.41 (2015–2019), 0.38 (2020–2021), 0.40 (2022–2026); correlations 0.06 and 0.30
- Interpretation: the correlation rose, but the smoothed R^2 is biased upwards and similar in both periods; only the 2020 island at 8–20 weeks is clearly significant: coupling is episodic, not a regime

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B8: CEE stock markets and the DAX by scale [Proposed]

- **Question**: how integrated are the Warsaw, Budapest and Prague markets with the DAX, horizon by horizon?
- **Inputs**: EODHD daily returns of WIG20, BUX, PX and DAX on common trading days, 2000–2026; MODWT LA(8), $J = 8$; model: A7 and B7
- **Tasks**
  1. Compute the wavelet correlation of each index with the DAX at levels 1–8 with 95% Fisher-z intervals.
  2. Compute the wavelet beta on the DAX at levels 1 and 8.
  3. Compare with the daily correlation.
  4. Interpretation: what do the scales add to the daily correlation for a long-horizon investor?
- **Report**: a chart, nine correlations, six betas, one sentence
- Notebook: section B8

C1: are CEE business cycles converging to the euro area? [Proposed]

- **Question**: has business-cycle synchronisation with the euro area increased for Bulgaria, Czechia, Hungary, Poland, Romania and Slovakia, once common extreme shocks are accounted for?
- **Inputs**: Eurostat quarterly GDP growth, 1995–2026; dynamic correlation over 6–32 quarters; model: B3 and the dynamic correlation of the notebook
- **Tasks**
  1. Write a five-line pre-registration: hypothesis, countries, subsamples, the treatment of 2020–2021, what would change your mind.
  2. Compute the band dynamic correlation for 1996–2012, 2013–2026 and 2013–2019.
  3. Interpretation: is the increase robust to excluding the pandemic?
- **Report**: a five-line plan, a table of six countries and three periods, a project plan
- Notebook: section C1

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C2: audit an AI answer [Proposed]

- A student asked an AI assistant to summarise spectral and wavelet methods. The answer:
- (a) The periodogram is a consistent estimator of the spectrum because it is asymptotically unbiased.
- (b) The HP filter with lambda = 1600 is a band-pass filter for cycles of 6 to 32 quarters.
- (c) A squared coherence of 0.30 estimated with 7 tapers is significant at 5%.
- (d) A phase of pi/2 at a period of 8 quarters means that the first series leads by 2 quarters.
- (e) The MODWT, like the DWT, needs a sample size that is a power of two.
- (f) Wavelet coherence can be computed without smoothing; the smoothing only makes the picture nicer.
- (g) For the Morlet wavelet with w0 = 6 the Fourier period equals the scale.
- (h) A Breitung-Candelon test in a VAR(2) can find causality at some frequencies and not at others.
- Tasks
  - 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (notebook section C2).
  - 2. Report: a list of eight verdicts with one line of justification each.

Key takeaways

- Spectral bands are χ^2 bands with equivalent degrees of freedom; they ignore bias near peaks
- Coherence, phase and gain answer different questions; test coherence before reading phase
- The HP filter makes cycles from random walks; check filters by their gain and in real time
- Wavelets give co-movement by scale and in time; significance needs a Monte Carlo null
- An AI answer is a draft: check conventions, thresholds and the scale–period factor

After the seminar

- Lecture 11 develops each topic of today: estimation theory and multitaper, cross-spectra and causality, filters and Hamilton's critique, wavelets and wavelet coherence
- C1 can grow into a team project: business-cycle convergence of Central and Eastern Europe with the euro area
- Reading: Thomson (1982); Croux, Forni and Reichlin (2001); Breitung and Candelon (2006); Hamilton (2018); Torrence and Compo (1998)
- **The seminar is for practice and is not graded**

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